

Project 3

In the following exercise, we will perform linear regression to fit various data sets and to predict outputs. Perform the following analyses by starting a new notebook.

Part A

First, we will use the Ch3PartA dataset to generate polynomial regressions using scikit learn. This dataset contains 100 observations of points x and their corresponding response, y . The data is divided into a training set (x_{tr}, y_{tr}) and a test set (x_{te}, y_{te}) .

1. Load **Ch3PartA.csv** into your notebook.
2. Create a scatter plot of: (a) y_{tr} against x_{tr} and another of (b) y_{te} against x_{te} . Notice the similarities and differences between the plots. **Answer the Question: What is the maximum value of y in the training set? In the test set?**
3. Generate the necessary features to fit polynomial regressions up to the 20th degree (up to and including the x^{20} term) on the training data. *Hint:* You will be fitting multivariate linear regression models with polynomial features of x . Familiarize yourself with *sklearn.preprocessing.PolynomialFeatures*.
4. Calculate the training MSE and the test MSE for 20 polynomial models up to degree 20. Store these as lists named *mse_train* and *mse_test* respectively. *Hint:* Familiarize yourself with *sklearn.metrics.mean_squared_error* and try to automate the process.
5. Generate a plot of both the training MSE and test MSE against flexibility for degrees 1 to 20. **Answer the Question: What is the minimum training MSE? Test MSE?**
6. From your plot, make an educated guess about the polynomial degree of the function that was used to generate the data. Then, give an estimate of the irreducible error $\text{Var}(\epsilon)$ for the optimal model on both the training set and test set. *Hint:* Revisit the section on hypothesis testing. Think about the relationship between MSE, RSS, and RSE. **Enter your answers.**

Part B

Next, we will use the Ch3PartB dataset to observe the effects of collinearity using statsmodels. This dataset contains 100 observations of points x_1 , x_2 , and y , the response variable.

1. Load the data from **Ch3PartB.csv** into a pandas DataFrame.
2. Show a scatterplot displaying the relationship between x_1 and x_2 ? **Answer the Question: What is the correlation coefficient between x_1 and x_2 ?**
3. Using the data, fit a least squares regression to predict y using x_1 and x_2 . Describe your results. *Hint: Familiarize yourself with `statsmodels.formula.api.ols`.*

Answer the Question: What are the estimates $\hat{\beta}_0$, $\hat{\beta}_1$, and $\hat{\beta}_2$? At a 95% confidence level, can you reject the null hypothesis $H_0 : \beta_1 = 0$? What about $H_0 : \beta_2 = 0$?

4. Now fit a least squares regression to predict y using only x_1 . Comment on your results. **Answer the Question: Can you reject the null hypothesis $H_0 : \beta_1 = 0$?**
5. Fit a least squares regression to predict y using only x_2 . Comment on your results. **Answer the Question: Can you reject the null hypothesis $H_0 : \beta_2 = 0$?**
6. Do Part B Questions 3-5 contradict each other? Explain why.